

- 5 (a)** e.g. 'storage of charge' / storage of energy
 blocking of direct current
 producing of electrical oscillations
 smoothing
 (any two, 1 mark each) B2 [2]
- (b) (i)** capacitance of parallel combination = $60\ \mu\text{F}$
 total capacitance = $20\ \mu\text{F}$ C1
A1 [2]
- (ii)** p.d. across parallel combination = $\frac{1}{2} \times$ p.d. across single capacitor
 maximum is 9V C1
A1 [2]
- (c)** *either* energy = $\frac{1}{2}CV^2$ *or* energy = $\frac{1}{2}QV$ and $Q = CV$ C1
 energy = $\frac{1}{2} \times 4700 \times 10^{-6} \times (18^2 - 12^2)$ C1
 = 0.42 J A1 [3]

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6 (a) (i) straight line with positive gradient

M1